

Spring 26 CSCI 240 -- Review for Exam 2

Textbook

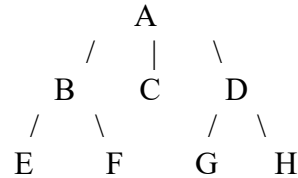
- Chapter 8
- Chapter 9
- Chapter 10
- Chapter 11

Review

- HW 2
- PAs 5 to 8
- In-class exercises and pop quizzes
- Trees – general trees, binary trees, arithmetic expression trees, tree representations, different traversals, depth and height, some tree algorithms
- Binary trees -- data representations (linked structure and array/vector), proper binary trees, traversals, depth, and height
- Priority Queues -- the PQ ADT, applications, implementations, comparators
- Heaps -- properties, vector/array representation, up-heap bubbling and down-heap bubbling, heap sort
- The Map ADT -- common operations, simple implementations, and applications, unordered maps and ordered maps
- Hashing -- hash functions (hash codes, esp. for strings, and compression), collision-handling schemes (separate chaining and open addressing with linear probing and double hashing), load factors and table size, clusters, performance
- Skip Lists -- representation, operations, expected performance
- Binary Search Trees -- properties, operations, and implementation
- Other Search Trees -- able to work with AVL trees, basic understanding of 2-4 trees and splay trees
- Problems – select appropriate data structures, use one ADT to implement another ADT

Some Review Questions/Exercises

1. Provide the pseudocode/code that performs a certain traversal on a binary tree – both recursive and non-recursive approaches.
2. Given a tree, perform 3 different traversals.



3. Given a binary tree (remove node C from the tree above, perform 4 different traversals.
4. Given a node in a tree that has the root as its grandparent, what is its depth?
5. Define the height of a tree and provide an efficient solution to compute the height of a tree.
6. Select an appropriate data structure for a certain situation – e.g., maintain a line where VIP gets to do first, display a list of values in a certain range
7. What is a PQ and what does it has to do with a heap?
8. What is a complete binary tree? A full binary tree?
9. Insert items into a PQ implemented with a heap and remove some items
10. Given a PQ, use it to sort a list of values and determine its runtime.
11. What are some issues that affect hashing performance?
12. If n is 7 and load factor is 0.5, what should the ideal table size be?
13. Insert a few values into a given hash table using a hashing scheme
14. Describe and draw a possible skip list with 5 items – 6 3 9 4 5
15. Insert items into a BST and remove some items
16. Insert items into an AVL Tree (10, 6, 8, 15, 20) and remove some items (10, 6)
17. Describe 2-4 trees and splay trees
18. Given a text file, provide an efficient solution to find the number of unique words.
19. Given a text file, provide an efficient solution to find the word that occurs the most often.
20. What is a comparator? How do you set one up in either C++ or Java?